

1. Introduction

This document is a guided tour of `mos-lsp`, the Mosaic language server. It is also an ordinary document: the headings are numbered and the prose cross-links to them. Section 2 lists what the server can do, section 3 covers installation, and the closing remarks sit in section 4 on page 2.

Each cross-reference above prints as a number in the PDF, but in the `.mos` source it is a live `@label` reference. That duality is the whole point: readers of the PDF get clean section numbers, while an editor wired to `mos-lsp` can navigate and rename the same references in the source.

2. Language server features

`mos-lsp` ships three capabilities today:

- Diagnostics surface compiler errors as you edit.
- Go-to-definition jumps from a reference to the heading that declares its label.
- Rename rewrites a label's declaration and every reference together.

Citations resolve too: the nod to the classic typesetting manual [1] and the bibliography-style reference [2] print as numbered markers.

2.1. Go-to-definition

Go-to-definition jumps from a reference to the heading that declares its label. This section demonstrates it on itself: it holds a live reference to its own heading, section 2.1, which the PDF prints as the number you see. In the source that token is `@gotodef`; put the cursor on it and run `go-to-definition` to land on this heading's `<gotodef>` label.

2.2. Rename

Rename rewrites a label's declaration and every reference to it in one edit. This heading declares the `<rename>` label, and the cross-link here, section 2.2, is a reference to it. Rename either the declaration or this reference, and both change together, in view, without scrolling away from this section.

2.3. Diagnostics

To watch diagnostics appear, introduce an error in the source: point a reference at a label that does not exist, or cite an unknown bibliography key. `mos-lsp` reports the problem inline; `undo` to clear it.

2.4. Figure labels

Figures carry labels too, through the `label:` argument:



Figure 1: A labelled figure you can reach with go-to-definition.

The figure above is labelled `diagram`, and this sentence references it as Figure 1 (which the PDF prints as its Figure N number). Go-to-definition on the `@diagram` token jumps to the figure, and `rename` rewrites the figure's label and every reference together, exactly as for heading labels.

3. Setup

Install the server so it lands on PATH: run `cargo mosi ls` from the workspace, then open this project in your editor. Section 1 has the overview and section 4 has the final check.

4. Conclusion

If go-to-definition reached this heading from a reference, `rename` swept a label cleanly, and the page printed for section 4 matches where you are reading, the language server is wired correctly.

5. References

The works cited above render below in first-use citation order; the entry list is produced by the `#bibliography` declaration that follows this paragraph.

- [1] Knuth, Donald E. The TeXbook. Addison-Wesley. 1984.
- [2] Patashnik, Oren. Designing BibTeX Styles. 1988.